MIL-D-2493
9 October 1950

MILITARY SPECIFICATION

DETONATORS, LOADING, ASSEMBLING, AND PACKING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-78-7
7 November 1946

This specification consists of this cover sheet and Specification 50-78-7, dated 7 November 1946, attached hereto, without modification.

Copies of this specification may be obtained upon application to the Office, Chief of Ordnance, Washington 25, D. C.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

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U. S. ARMY
SPECIFICATION

No. 50-78-7
7 November 1946

DETONATORS, LOADING, ASSEMBLING, AND PACKING

A. APPLICABLE DRAWINGS AND SPECIFICATIONS.

A-1. The Ordnance Department detonator assembly drawing listed below pertaining to the detonator under contract, and all drawings and specifications listed or referred to thereon, together with the following drawings and specification, and all drawings and specifications listed or referred to thereon, form a part of this specification:

82-3-282	73-2-186	73-2-190	73-2-205
82-3-337	73-2-187	73-2-191	73-2-211
73-2-184	73-2-188	73-2-199	73-2-231
73-2-185	73-2-189	73-2-204	73-2-245

81-3-150 Fixture, Testing, Detonator Sensitivity
81-3-64 Bomb, Sand Test for Detonators
Federal Specification RR-S-366, Sieves; Standard Testing

A-2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPES.

B-1. This specification covers the following detonators:

M13	M17	M19A2	M21	M23	M28	M30	M35
M15A2	M18	M20	M22	M24	M29	M31	M37

C. MATERIAL AND WORKMANSHIP.

C-1. Materials and parts shall be in accordance with applicable specifications and drawings.

C-2. All parts shall be free from chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

D. GENERAL REQUIREMENTS.

D-1. Lead azide. The moisture content of the lead azide shall not exceed 0.50 percent. (See par. F-3.)

D-2. Primer mixture. The moisture content of the primer mixture shall not exceed 0.30 percent. (See par. F-5a and par. F-6a.)

E. DETAIL REQUIREMENTS.

E-1. The detonators shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. Igniting mixture. The moisture content of the igniting mixture shall not exceed 0.30 percent. (See par. F-7a.)

E-3. Detonator assembly.

E-3a. When sealing is prescribed the detonator shall be waterproof, as evidenced by normal functioning after water immersion when tested in accordance with paragraph F-8e.

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E-3b. Table I specifies the detonators, and requirements, for test as prescribed in paragraphs F-8a and F-8b. Each detonator assembly shall function on the first attempt and shall crush not less than the minimum amount of sand indicated.

TABLE I

DETONATOR ASSEMBLY	MINIMUM AMOUNT OF SAND CRUSHED
M17	20 grams
M21	17 grams
M30	10 grams
M31	45 grams
M35	34 grams

E-3c. Sensitivity and functioning requirements of stab actuated detonators are as follows:

E-3c(1). Table II specifies the detonators, and requirements, for test as prescribed in paragraph F-8c. The detonator shall function on the first attempt and shall perforate the lead disc to not less than the minimum diameter indicated.

TABLE II

STANDARD STEEL BALL (See par. H-2)

DETONATOR ASSEMBLY	DIAMETER (Inches)	WEIGHT (Ounces)	HEIGHT OF DROP (Inches)	MIN. DIAMETER OF PERFORATED HOLE IN LEAD DISC (Inches)
M13	1-3/16	3.95	12	1/4
M15A2	1-3/16	3.95	3	3/8
M18	1-3/16	3.95	12	3/16
M19A2	1-3/16	3.95	3	3/8
M20	1-3/16	3.95	12	5/32
M22	1-3/16	3.95	12	1/8
M23	1-7/8	15.55	23	1/8
M24	1-3/16	3.95	12	3/32
M28	1-3/16	3.95	12	5/16
M29	1-3/16	3.95	12	1/16
M37	1-3/16	3.95	12	3/16

E-3c(2). Acceptability will be determined in accordance with Table III. If at any time in the course of testing the number of detonators shown in Column 3 the number of failures equals the number shown in Column 1, testing will be discontinued and the lot rejected. If the total number tested reaches the number shown in Column 2 with a number of failures not exceeding that shown in Column 1, the lot will be accepted. The order of testing detonators from production units shall be in rotation, so that approximately equal proportions from each production unit will have been tested at any given time.

TABLE III

Column 1	Column 2	Column 3
0	300	
1	550	
2	800	
3	1050	250
4	1300	500
5	1550	750
6	1800	1000
7		1800

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(NOTE: The failure of the detonator either to function on the first attempt or to perforate the lead disc to the required diameter will be considered as a "f-'ure" under Column 1 of Table III.)

E-3c(3). If three or more failures occur in tests of detonator assemblies from any one prodcution unit, and if the lot undergoing test is acceptable under Table III, that production unit shall be withdrawn from the lot and scrapped without cost to the contractor. When this procedure is followed, the acceptability of the lot is not affected.

E-4. The detonator assemblies shall contain no defects which can be observed by visual inspection.

E-5. Check test for possible deterioration. If the total elapsed time between original acceptance of any detonator lot and assembly of that lot into the ammunition component with which it is to be issued exceeds two months or if the detonators have been subjected to adverse conditions at any time since previous tests, the detonator lot shall be subjected to, and must satisfactorily pass, the check test outlined in paragraph F-8d immediately before the lot is used. (This test does not normally apply to detonators used in subsequent assembly operations within two months of the original acceptance test.)

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. Inspection.

F-1a. Inspection shall be as specified in U. S. Army General Specification for Ammunition Except Small Arms Ammunition, No. 50-0-1.

F-1b. Visual examination. Such visual inspection shall be made as is necessary to insure absence of defective assemblies.

F-1c. Sealing. When sealing is prescribed, such inspection shall be made as is necessary to insure that the sealing operations are performed properly.

F-1d. Primer mixture and igniting mixture. The inspector shall ascertain that a suitable system of control testing is maintained by the contractor to insure, prior to releasing batches of primer mixture and igniting mixture for production loading, that the sensitivity and other applicable requirements are met.

F-2. Size of lots. One hundred thousand maximum. A smaller quantity may constitute a lot at the discretion of the inspector. Two thousand plus or minus 500 shall constitute the first lot manufactured on any contract, except when a new contract immediately follows the manufacture of satisfactory material of the same character. Each lot shall contain metal parts manufactured in accordance with one drawing and revision. The detonator lot shall be so controlled that all detonators produced by any one production unit during one shift may be readily segregated at any time prior to acceptance. (This may be accomplished in any convenient manner which insures easy identification without marking on, or otherwise changing the appearance of the detonator itself.)

F-3. Determination of moisture content of lead azide. If lead azide is used in loading the detonators, the moisture content of the lead azide shall be determined at intervals as prescribed by the inspector. When such determination is made, a 2-gram sample of lead azide shall be dried at 95°C. for 1.5 hours. The loss in weight shall be calculated as the percentage of moisture. (See par. D-1.)

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F-4. Determination of graphite content of tetryl. If tetryl is used in loading the detonators, the inspector shall assure himself that the graphite content of the tetryl does not exceed the amount prescribed on the drawing. If the inspector decides that a test is necessary, the tetryl shall be analyzed as follows:

F-4a. Weigh accurately a sample of approximately 5 grams of the tetryl and transfer to a 250 ml beaker. Add 100 ml of acetone and heat to dissolve the tetryl. Filter the solution through a tared filtering crucible and wash the residue in the crucible with two 25 ml portions of acetone. Dry the crucible at 100°C for one hour; cool in a desiccator, and weigh. Calculate the increase in weight as the percentage of graphite.

F-4b. Retest. If the analysis indicates that the graphite content does not exceed the amount prescribed on the drawing by more than 0.25 percent, the tetryl may be reblended and resubmitted for test.

F-5. Test of primer mixture. (Applicable to the M15A2 and other detonator assemblies containing the same primer mixture). The moisture content of the primer mixture shall be determined and the ingredients shall be verified at intervals as prescribed by the inspector. When such tests are made, the primer mixture shall be analyzed as described in the following paragraphs.

F-5a. Moisture content. Dry to constant weight at 55° to 65°C, or in a desiccator containing calcium chloride, a known weight, 0.40 to 0.70 gram, of the primer mixture. Calculate the loss in weight as the percentage of moisture.

F-5b. Antimony sulphide. Transfer a weighed portion of approximately 1 gram of the sample to a beaker containing 100 ml of 1 percent nitric acid. Allow the mixture to digest for 10 minutes at room temperature with occasional agitation. Filter off the insoluble matter on a tared filtering crucible. Wash this residue with distilled water. Retain the filtrate and washings for the determination of lead sulphocyanate content. Rinse the residue with alcohol and ether, dry at approximately 70°C for 1 hour, cool in a desiccator, and weigh. Multiply the gain in weight by 1.01 for this composition and calculate as percentage of antimony sulphide in the sample on a moisture-free basis.

F-5c. Lead sulphocyanate. Dilute the combined filtrate and aqueous washings from the antimony sulphide determination to approximately 300 ml with distilled water. Add 15 ml of 10 percent silver nitrate solution while agitating the solution vigorously. Continue the agitation until the precipitate coagulates. Allow the precipitate to settle for about 10 minutes, filter, catching the precipitate on a tared filtering crucible having a thick asbestos mat. Wash the residue with distilled water. Apply light suction to the filtering crucible assembled to a clean suction flask and treat with 25 ml of 3 percent ceric ammonium nitrate solution, keeping the crucible covered with a watch glass to retain spatterings due to gas evolution. Keep the suction adjusted so that the 25 ml of solution passes through the crucible in about 5 minutes. Repeat this extraction treatment, using the same ceric ammonium nitrate solution and keeping the crucible covered until the spattering ceases. Wash the material from the watch glass into the crucible with distilled water. Using a rubber policeman assembled on a glass rod, tamp the precipitate in the crucible into a compact pad which does not crack

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when air is drawn through it. Continue treating the residue in the crucible with the same ceric ammonium nitrate solution in the same manner, keeping the residue tamped compactly, until no further evolution of gas is detected by careful observation. (A total of six treatments is usually required.) Then repeat the procedure 3 times subsequent to the last treatment which causes evolution of gas. Be sure to return to the crucible any asbestos which may have been removed during the above-mentioned treatments. Wash the residual silver sulphocyanate with distilled water, rinse with alcohol and ether, aspirate until free of ether, dry at approximately 135°C for one hour, cool in a desiccator, and weigh. Calculate the weight of residual silver sulphocyanate to percent of lead sulphocyanate on a moisture-free basis as follows:

$$\text{Lead sulphocyanate, percent} = \frac{97.4A \times 0.985}{W\,(100 - B)}$$

Where:

A = weight of residual silver sulphocyanate
B = percentage of moisture
W = weight of sample.

F-5d. Lead azide. Determine the lead azide content using the apparatus shown assembled in Fig. 1 and maintaining it at as near constant temperature as is practicable. Accomplish this by protecting the gas system from the effects of air currents or bodies at appreciably different temperatures. Transfer a weighed portion of approximately 2 grams of the sample to a 125 ml Erlenmeyer flask (D), add 5 ml of 10 percent silver nitrate solution and 10 ml of distilled water. Shake the mixture and wash down the sides of the flask with distilled water. Fill the delivery burette (A) with 60 percent ceric ammonium nitrate solution. Connect the Erlenmeyer flask (D) tightly to the gas burette (E) (a gas burette calibrated in 0.1 ml and having a total capacity of 100 ml for the calibrated portion is satisfactory) by means of a tightly fitting rubber stopper (C), and clamp this assembly in position. Open stopcock (F) and using the leveling bulb (G) adjust the water level in the gas burette (E) to the zero point. Close the stopcock (F) and test the system for gas-tightness by raising and lowering the leveling bulb (G) and releveling the water in the bulb (G) with that in the burette (E). Consider the system to be gas-tight if the water level in the burette (E) returns to the zero point. Add 5.0 ml of the 60 percent ceric ammonium nitrate solution from the delivery burette (A), lower the leveling bulb (G) to the vicinity of the 30 ml mark on the gas burette (E), and shake the reaction flask (D) very vigorously, until no further increase in gas volume occurs. (This may conveniently be accomplished by agitation of the entire assembly.) Transfer to a 400 ml beaker (H) approximately 200 ml of water at the temperature of the gas burette as shown by thermometer (I), and raise the beaker under the reaction flask (D) until the water level rises about 2 inches above the bottom of the flask. Clamp the beaker in position and allow the apparatus to stand with occasional shaking until the temperature of the gas in reaction flask (D), as indicated by thermometer (B), is the same as that of the gas in the gas burette (E), as indicated by thermometer (I), and until no further change in gas volume occurs. Adjust the height of the leveling bulb (G) so

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that the level of the water in it is the same as the level of the water in the gas burette (E). Record the observed volume in the gas burette (E), the temperature of the gas, and the atmospheric pressure. Calculate the percentage of lead azide in the sample on a moisture-free basis as follows:

$$\text{Lead azide, percent} = \frac{(A - 5.0)(B - C)\ 0.0570 \times 1.08}{W(100 - D)(1 + 0.00367t)}$$

Where:

A = gas burette reading, ml
B = atmospheric pressure, mm of mercury*
C = vapor pressure of water at t°C
D = percentage of moisture in the sample
t = temperature of gas burette and reaction flask, °C.
W = weight of sample, grams.

*(Note. If the pressure is observed on a mercurial barometer having a brass scale, calculate the corrected atmospheric pressure as follows:

$$\text{Atmospheric pressure (B), mm} = E - 0.000163EF$$

Where:

E = reading in mm of mercurial barometer
F = temperature of barometer in °C).

F-5e. Potassium chlorate. Calculate the percentage of potassium chlorate in the sample on a moisture-free basis by subtracting from 100 the sum of the percentages of antimony sulphide, lead sulphocyanate, and lead azide.

F-6. Test of primer mixture. (Applicable to the M13 and other detonator assemblies containing the same primer mixture). The moisture content of the primer mixture shall be determined and the ingredients shall be verified at intervals as prescribed by the inspector. When such tests are made, the primer mixture shall be analyzed as described in the following paragraphs.

F-6a. Moisture content. Dry to constant weight at 55° to 65°C, or in a desiccator containing calcium chloride, a known weight, 0.40 to 0.70 gram of the primer mixture. Calculate the loss in weight as the percentage of moisture.

F-6b. Potassium chlorate. Weigh accurately approximately 1 gram of the primer mixture and transfer to a tared ignited filtering crucible of approximately 15 ml capacity. Add 3 ml of lead azide-saturated distilled water at a temperature of 25° ± 2°C. Agitate for exactly one minute, if necessary, breaking up any lumps with a glass rod, taking care not to disturb the asbestos mat. Apply suction. Repeat this procedure five times, making a total of six 3 ml extractions. Rinse the sample in the filtering crucible with a few ml of alcohol and ether; dry for approximately 15 minutes at 95°C; cool in a desiccator, and weigh. Calculate the loss in weight as the percentage of potassium chlorate in the sample.

F-6c. Lead azide. Extract the residue from the potassium chlorate determination with 5 ml portions of saturated ammonium acetate solution at about 25°C; (hot saturated ammonium acetate solution must not be used as it dissolves antimony sulphide to some extent) agitating each portion for about

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30 seconds, until the washings no longer give a yellow precipitate with a few drops of potassium dichromate solution. The use of more than 60 ml of ammonium acetate solution should be avoided. Wash the residue with distilled water; rinse with a few ml of alcohol and ether; dry for approximately 15 minutes at 100° ≠ 10°C; cool in a desiccator, and weigh. Calculate the loss in weight as the percentage of lead azide in the sample.

F-6d. Antimony sulphide. Extract the residue from the lead azide determination with hydrochloric acid until free from antimony sulphide. (This extraction can be made without removing the sample from the filtering crucible and thus running the risk of disturbing the asbestos mat. Apply suction and extract most of the antimony sulphide with cool concentrated hydrochloric acid, then rinse thoroughly with hot concentrated hydrochloric acid.) Rinse the crucible and residue thoroughly; ignite to remove separated sulphur and organic matter; cool in a desiccator, and weigh. Calculate the loss in weight as the percentage of antimony sulphide.

F-6e. Carborundum. Subtract the original weight of the ignited filtering crucible from the weight of the crucible and residue after the antimony sulphide determination and calculate the difference as the percentage of carborundum in the sample.

F-7. Test of igniting mixture. (Applicable to the M31 and other detonator assemblies containing the same igniting mixture.) The moisture content of the igniting mixture shall be determined and the ingredients shall be verified at intervals as prescribed by the inspector. When such tests are made, the igniting mixture shall be analyzed as described in the following paragraphs:

F-7a. Moisture content. Dry to constant weight at 55°C to 65°C, or in a desiccator containing calcium chloride, a known weight, 0.40 to 0.70 gram, of the igniting mixture. Calculate the loss in weight as the percentage of moisture.

F-7b. Lead sulphocyanate. Transfer a weighed portion of approximately 1 gram of the sample to a 250 ml beaker. Add 20 ml of distilled water and stir with a rubber policeman until the sample is thoroughly wetted. Add a known volume of approximately 35 ml of standardized N/10 silver nitrate solution. Stir and add 100 ml of 10 percent nitric acid solution. (Nitric acid should be free of oxides of nitrogen.) Thoroughly stir the mixture for 5 minutes, add 5 ml of a saturated solution of ferric ammonium sulphate and then titrate with standardized N/10 potassium sulphocyanate solution to the first permanent pale rose coloration. Calculate the percentage of lead sulphocyanate in the sample as follows:

$$\text{Percentage of lead sulphocyanate}^{a} = \frac{A \times B \times 16.25}{C}$$

Where:

A = Difference between the total amount of silver nitrate added and the silver nitrate equivalent to the potassium sulphocyanate used

B = Normality of the silver nitrate solution

C = Weight of moisture-free sample.

a This calculation is based on a lead sulphocyanate purity of 99.50 percent.

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F-7c. Potassium chlorate. Determine the potassium chlorate content by subtracting the percentage of lead sulphocyanate from 100.

F-8. Proof acceptance tests. The tests shall be conducted as follows: (See par. H-4.)

F-8a. Sand test. (Applicable to M17, M21, M30 and M35 detonator assemblies.) A composite sample consisting of two detonator assemblies representing each production unit during one shift, but not less than thirty detonator assemblies (small lots may require the sampling of more than two detonator assemblies representing each shift to form the minimum composite sample) selected by the inspector from each lot shall be tested as follows: (see par. E-3b.)

F-8a(1). Insert one of the detonators in cup (Fig. 2) with the sensitive end pointing upward. Put 0.15 gram of loose dry lead azide on top of the detonator. Prick one end of an 8-inch length (approximate) of black powder fuse with a pin to loosen the powder in order to insure good flash. Insert the pricked end of the fuse into the cup, in such a manner that the end of the fuse will be in contact with the lead azide, and crimp it in place at the mouth end of the cup, care being taken to prevent the fuse from slipping out of position by marking the fuse and checking the position of the mark after crimping. Slip a small piece of rubber tubing, approximately the same diameter of the fuse so that it fits snug ly, over the opposite end of the fuse. Pour 80 grams weighed to an accuracy of 0.5 gram, of standard Ottawa sand, which passed through an 840 micron (No. 20) sieve, but was retained on a 590 micron (No. 30) sieve into a standard test bomb, Drawing 81-3-64. Strike vigorously with a hammer at least five times while rotating the bomb on its axis. Place the cup containing the detonator, lead azide, the fuse in the center of the bomb (Fig. 2). Pour into the bomb 120 grams of the standard Ottawa sand, sieved as above and weighed with an accuracy of 0.5 gram. Adjust the rubber tubing on the fuse to fit closely against the cover of the bomb, and secure the cover firmly. Strike the bomb vigorously at least five times while rotating the bomb on its axis. Fire the detonator by igniting the black powder fuse. Screen the sand by placing on a 590 micron (No. 30) sieve and shaking for three minutes, using a mechanical shaker. Weigh all the sand that passes through the sieve.

(Note. The sand used in this test shall be standard Ottawa sand which has not been used previously for a similar test. The sieves used shall be in accordance with Federal Specification RR-S-366, Sieves; Standard Testing.)

F-8a(2). Retest. A retest, using double the number of samples used in the original test, (samples selected in the same manner as the original sample), may be made, if requested by the contractor, provided not more than one detonator assembly fails to meet the requirements of paragraph E-3b.

F-8b. Sand test. (Applicable to the M31 detonator assembly.) A composite sample consisting of two detonator assemblies representing each production unit during one shift, but not fewer than thirty detonator assemblies (small lots may require the sampling of more than two detonator assemblies representing each production unit during one shift to form the minimum composite sample) selected by the inspector from each lot shall be tested as follows: (See par. E-3b.)

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F-8b(1). Insert one of the detonators in cup (Fig. 3) with the sensitive end pointing upward. Place a hollow black powder pellet (Fig. 3) on top of the detonator. Prick one end of an 8-inch length (approximate) of black powder fuse with a pin to loosen the powder in order to insure good flash. Insert the pricked end of fuse into the cup in such manner that the end of the fuse will be in contact with the black powder pellet, and crimp it in place at the mouth end of the cup, care being taken to prevent the fuse from slipping out of position by marking the fuse and checking the position of the mark after crimping. Slip a small piece of rubber tubing, approximately the same diameter of the fuse so that it fits snugly, over the opposite end of the fuse. Pour 80 grams, weighed to an accuracy of 0.5 gram, of standard Ottawa sand, which passed through an 840 micron (No. 20) sieve, but was retained on a 590 micron (No. 30) sieve, into standard test bomb, Drawing 81-3-64. Strike vigorously with hammer at least five times while rotating the bomb on its axis. Place the cup containing the detonator, pellet, and fuse in the center of the bomb (Fig. 3). Pour into the bomb 120 grams of the standard Ottawa sand, sieved as above and weighed with an accuracy of 0.5 gram. Adjust the rubber tubing on the fuse to fit closely against the cover of the bomb, and secure the cover firmly. Strike the bomb vigorously at least five times while rotating the bomb on its axis. Fire the detonator by igniting the black powder fuse. Screen the sand by placing on a 590 micron (No. 30) sieve and shaking for three minutes using a mechanical shaker. Weigh all the sand that passes through the sieve.

(Note. The sand used in this test shall be standard Ottawa sand which has not been used previously for a similar test. The sieves used shall be in accordance with Federal Specification RR-S-366, Sieves: Standard Testing.)

F-8b(2). Retest. A retest, using double the number of samples used in the original test (samples selected in the same manner as the original composite sample), may be made, if requested by the contractor, provided not more than one detonator assembly fails to meet the requirements of paragraph E-3b.

F-8c. Sensitivity test of detonator assembly. (Applicable to M13, M15A2, M18, M19A2, M20, M22, M23, M24, M28, M29 and M37 detonator assemblies.)

F-8c(1). Functioning test. A composite sample consisting of groups of detonator assemblies, each group proportionate in size to the number of assemblies from the corresponding production unit actually present in the lot, shall be selected at random by the inspector from each production unit shift, and shall be tested by assembling each detonator assembly as shown on Figure 4 (for all detonators except the M24 detonator) or as shown on Figure 5 (for the M24 detonator only), and testing in Fixture, Testing, Detonator Sensitivity, Drawing 81-3-150 (see par. E-3c). In determining the acceptability of a lot under Table III, tests conducted by the contractor may be considered, provided the inspector is satisfied that the samples have been so selected as to be representative of the lot, and that the method of testing complies with paragraph F-8c(1). When results of contractor's tests are considered, such results shall be restricted to the first 200 detonators fired from the lot, and all information pertinent to the functioning obtained with the quantity of detonators considered shall also be considered in determining acceptability of the lot.

(Note. The minimum hole diameter is that hole which will receive a plug gage equal to the minimum size given. A dead load of not more than five pounds should be applied to the gage in this determination to avoid false diameters resulting from burrs, etc.)

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F-8d. Check test for possible deterioration. This test is to be performed on samples selected by the Army Inspector of Ordnance at the plant assembling the detonator into the ammunition component with which it is issued. (See par. E-5.) Such tests shall be conducted at Government expense, without cost to the contractor loading the detonator or the contractor assembling the detonator into the item with which it is issued, and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted by either contractor while the lot is under such contractor's jurisdiction. (Note. When these tests are required, the detonator net lot size will be reduced by the number of assemblies expended in the applicable test procedure.) Should any lot of detonators fail the retest permitted in this phase, it shall be rejected for loading purposes subject to the disposition outlined in paragraph H-3. For the purposes of this test, the specified minimum number of samples shall be selected at random from each lot by the inspector and tested in accordance with paragraphs F-8a, F-8b, or F-8c, as applicable (including subparagraphs), using the information contained in these paragraphs as a basis of acceptability. When testing in accordance with paragraph F-8c, the lot will be rejected for loading only if the number of failures obtained with the minimum sample size indicated under Column 3 of Table III is sufficient to require rejection.

F-8e. Waterproofness test. When sealing is prescribed the effectiveness of the seal shall be tested as follows:

F-8e(1). Twenty-five detonator assemblies selected by the inspector from each lot shall be subjected to the waterproofness test. Immerse the sample detonators in water to a depth of not less than 2 inches nor more than 3 inches, after bringing both the detonators and the water to the same temperature, and allow to stand for a period of not less than 48 hours. Within one hour after removal from the water, test the detonators according to the applicable testing procedure to determine that the requirements of paragraph E-3 are met. No detonator shall fail in this test.

F-8e(2). Retest. A retest, using double the number of samples used in the original test, may be made, at the request of the contractor, providing not more than one detonator fails to function satisfactorily in the original test. One failure will be permitted in the combined results of the original test and the retest.

G. PACKAGING, PACKING, AND MARKING.

G-1. Packaging. As specified on the drawing.

G-2. Packing.

G-2a. As specified on the drawing.

G-2b. Data cards. Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted detonators to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards, shall be in accordance with instructions issued by the Office, Chief of Ordnance.

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AMMUNITION DATA CARD

Kind: _____ Ammunition Lot No. _____
Quantity in Lot _____
Drg. No.* _____ Spec. No.* _____
Metal parts manufactured by _____ Lot No. _____
Loaded and assembled by _____ Date _____
Contract No. _____ Order No. _____
Packed _____ in a box.
Remarks: _____

Certified to by _____ Inspector.
(Print here name of inspector, district, Arsenal, or loading plant, and subdistrict where applicable.)

*Include revision, amendments, and ANR's.

G-3. Marking. The marking for shipment shall be as prescribed on the drawing.

H. NOTES.

H-1. The detonators covered by this specification are intended for use in various types of ammunition.

H-2. Commercial steel balls can be purchased which will meet the requirements for the drop weights used in the sensitivity test.

H-3. Rejected lots. This specification is intended to provide suitable acceptance tests for detonators of normal quality. It is realized that abnormal cases will arise, particularly with respect to paragraph F-8d and special detonator requirements imposed by fuze specifications, which cannot be fully covered herein. Such abnormal cases, including final rejections chargeable to detonator loading contractors and not covered below, should be referred to the Office, Chief of Ordnance, Washington 25, D. C., together with all pertinent information, for review and recommendation of appropriate disposition. (When testing detonators for acceptability under paragraph E-3c, removal of unsatisfactory portions of the lot, except as permitted by paragraph E-3c(3), invalidates the test results previously obtained. The lot may, however, be resubmitted to the entire test, at the request of the contractor.)

H-4. Detonator assemblies which fail to function in tests required by this specification should be sectioned and examined, wherever practicable, to determine if there are abnormal conditions present which might lead to malfunctioning. This examination is for information only, and will not affect acceptability.

H-5. This specification replaces Picatinny Arsenal Tentative Specification PXS-1048.

H-6. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

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Note: Copies of this specification may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

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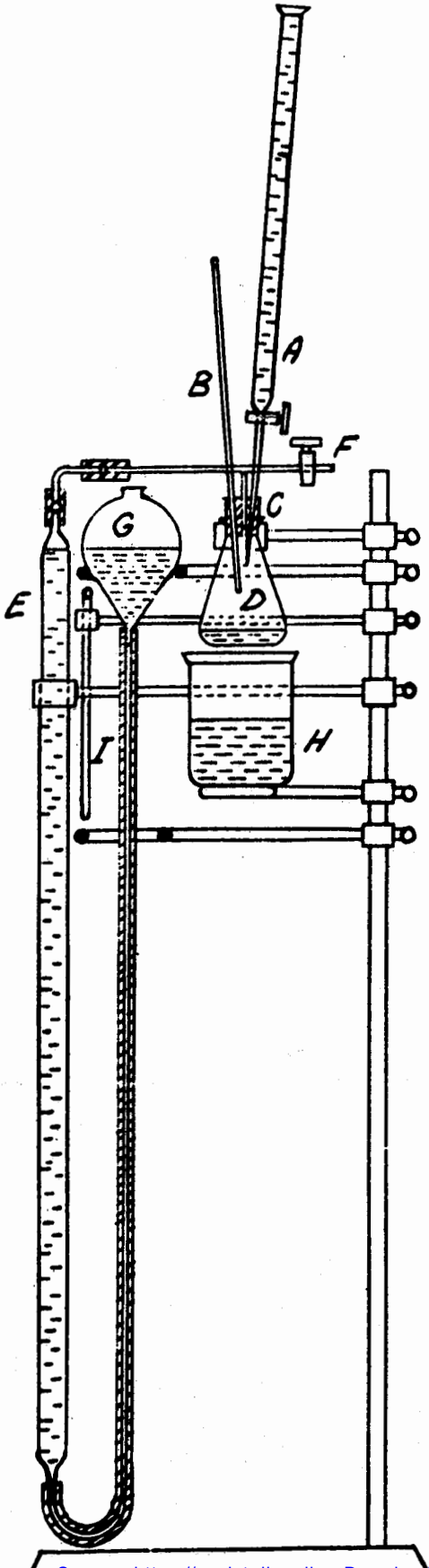
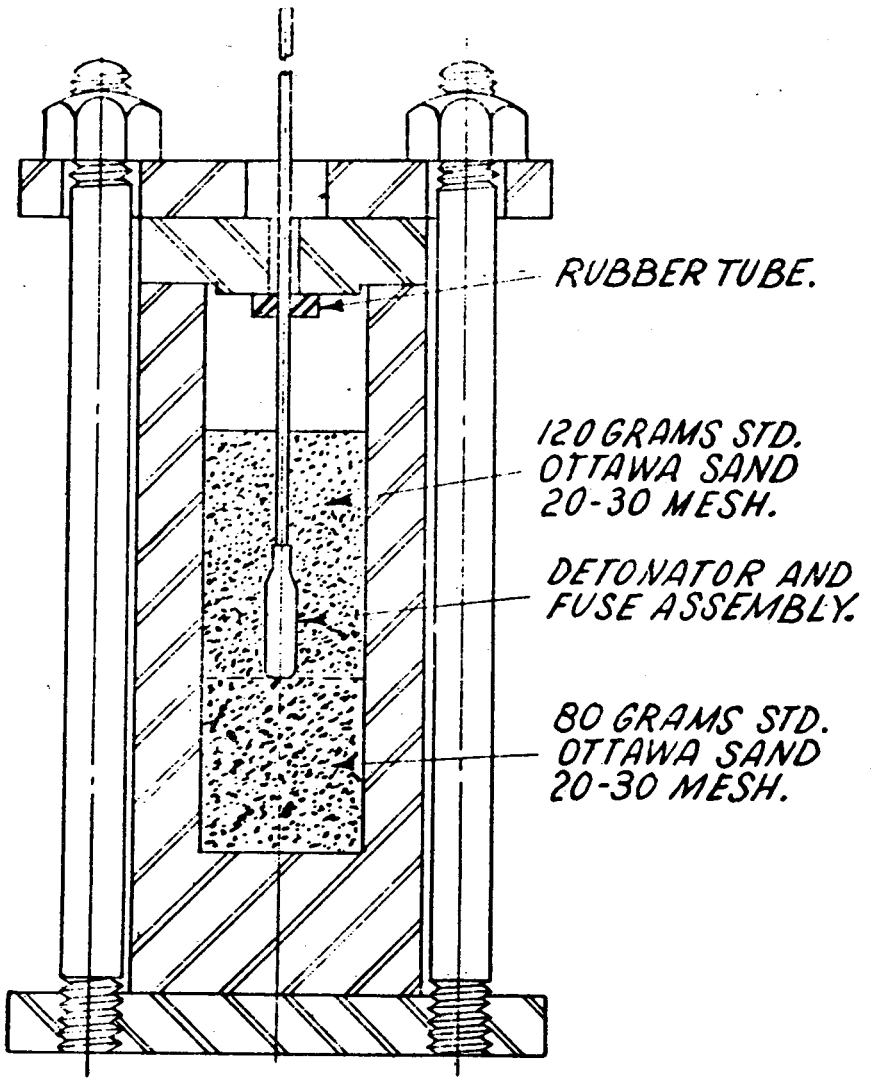
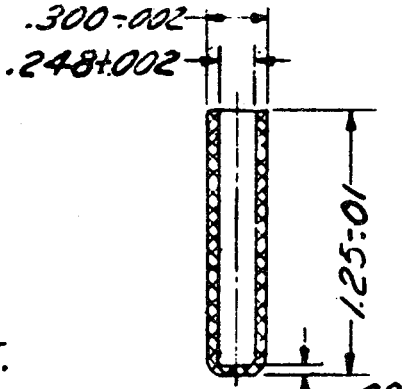


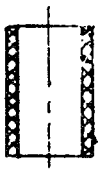
FIG. 1

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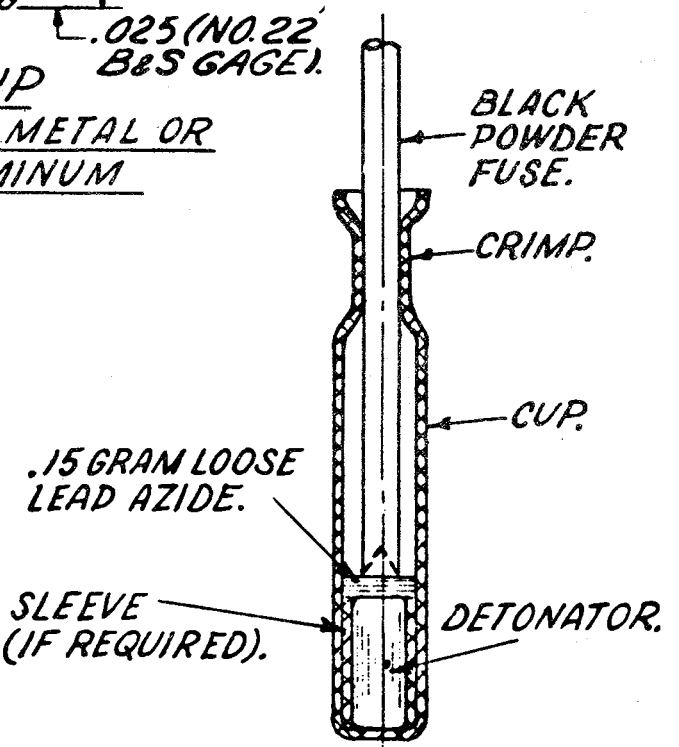




CUP
GILDING METAL OR ALUMINUM



SLEEVE
ALUMINUM

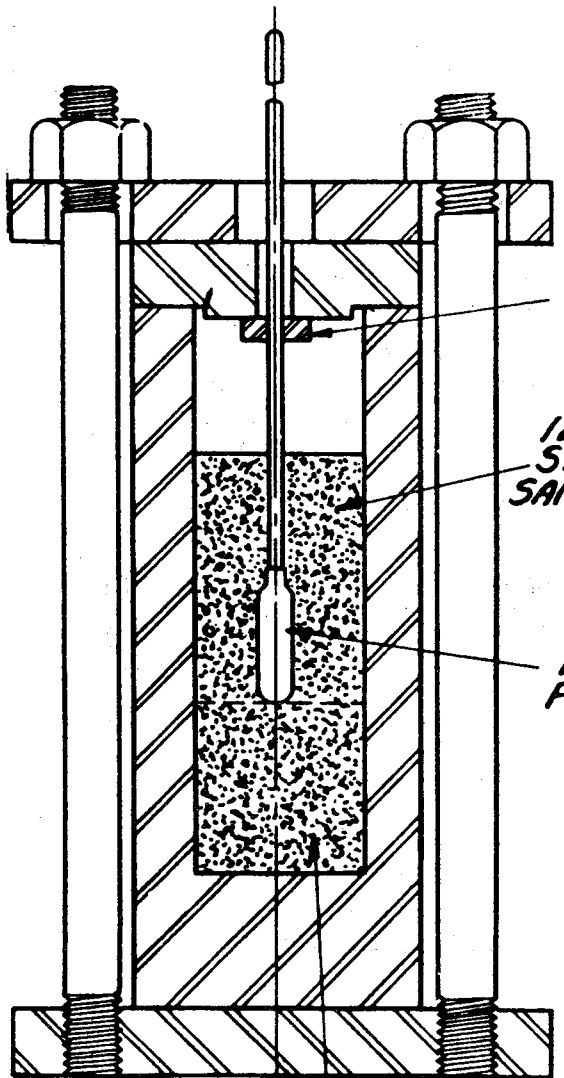


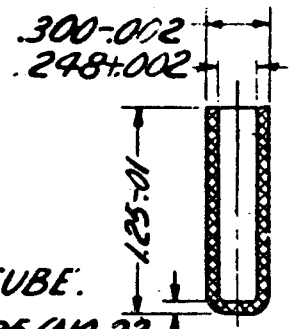
DETONATOR AND FUSE ASSEMBLY

BOMB SHOWN ON DRG. 81-3-64.

FIG 2

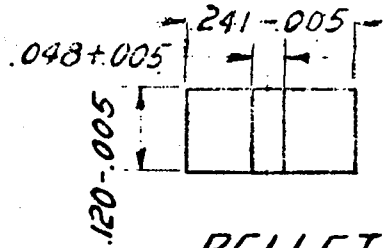
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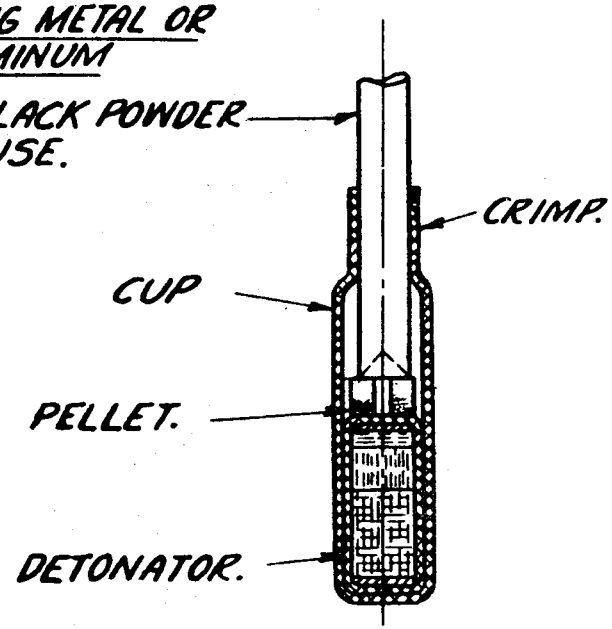
CUP

GILDING METAL OR
ALUMINUM



PELLET

2.39-.17 GRAINS (155-11 MILLIGRAMS)
ARMY BLACK POWDER, GRADE
"A-5", SPEC. 50-14-1
ADVISORY SPECIFIC GRAVITY 1.80



DETONATOR AND
FUSE ASSEMBLY

FIG. 3

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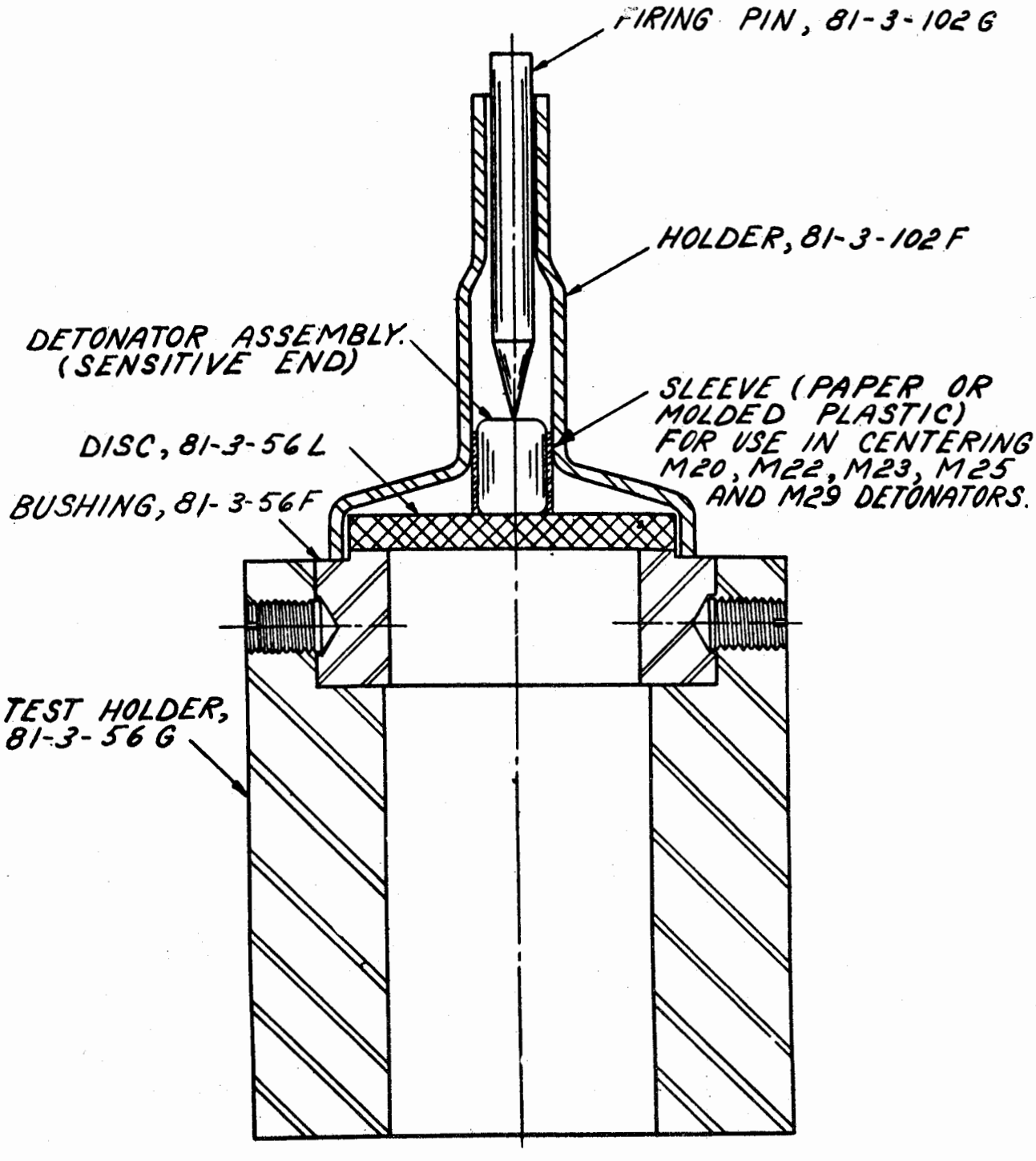


FIG. 4

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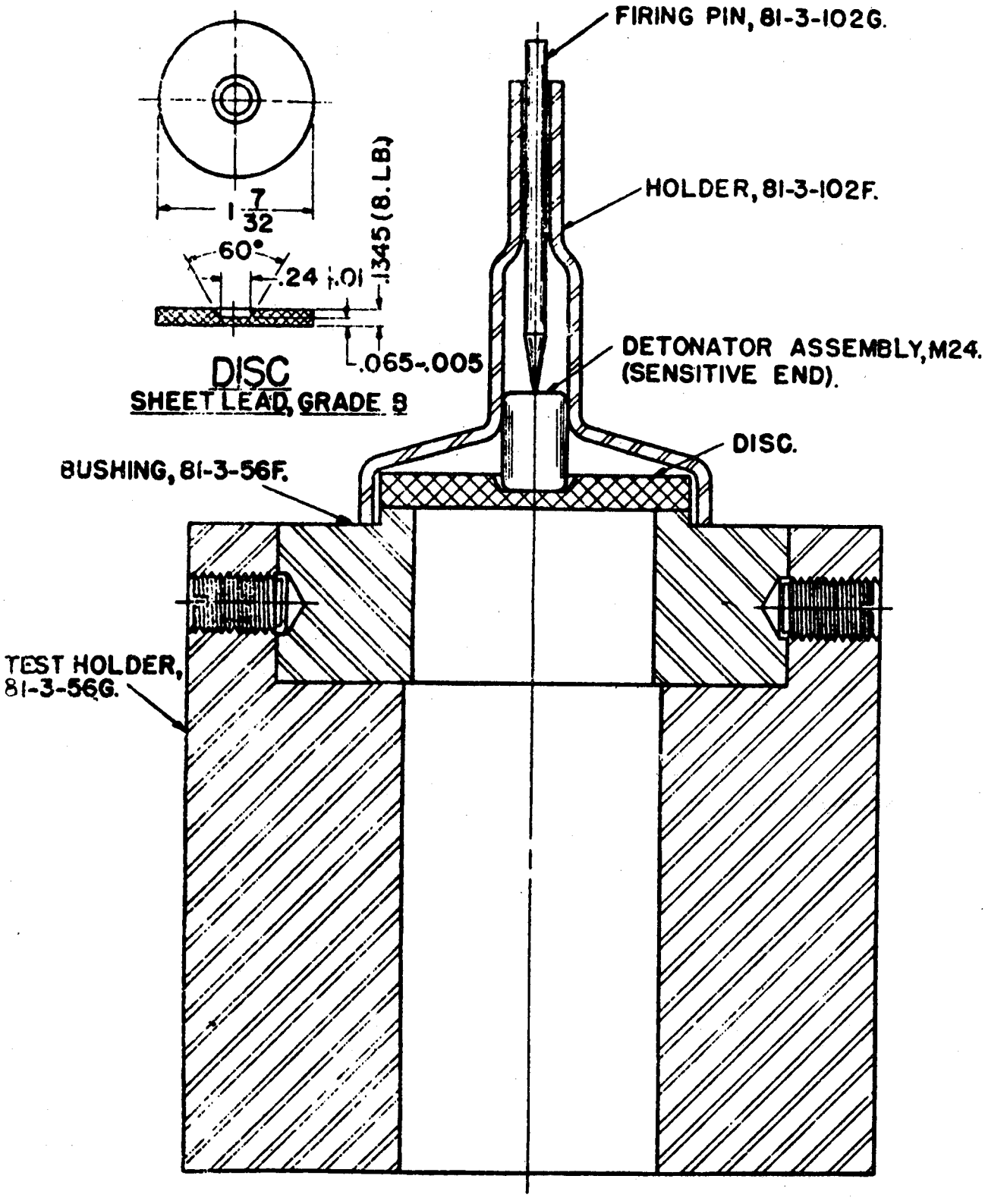


FIG. 5

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